CSE 803: Computer Vision Fall 2026

Department of Computer Science and Engineering
Michigan State University

Part 1: Course Information

Instructor: Zijun Cui cuizijun@msu.edu

Teaching Modality: In-Person

Office Hours: Thursday 1-2pm at EB 2212 or via Zoom

Course: Class is Tue/Thu 10:20 – 11:40AM
Location: **Hubbard Hall G29**

Prerequisites: programming, algorithms, data structures, probability and statistics, linear algebra, 3D calc.

Description: visual information processing problems; human and machine vision systems; image formation; image transforms, encoding, enhancement; edge detection; segmentation; 2D and 3D object description and recognition; scene analysis; applications

Objectives: to study problems associated with image acquisition, processing, and interpretation; to learn tools for solving computer vision problems in industry and various scientific disciplines; to understand the general process of designing a practical computer vision system.

Text: R. Szeliski. *Computer Vision: Algorithms and Applications.* Free copy@ http://szeliski.org/Book/;

References: D. A. Forsyth and J. Ponce. *Computer Vision: A Modern Approach (2nd Edition).*

Discussion: Piazza at https://piazza.com/msu/fall2026/cse803/home

Course Requirements

- Internet connection (DSL, LAN, or cable connection desirable)
- Access to Desire2Learn (D2L), or other delivery platform.
- Ability to scan and upload documents (e.g. CamScanner or Adobe Scanner app)
- Any other tools, resources, and materials needed by the student for the course.

If you do not have the necessary technology and/or equipment to adequately address the above requirements, the College of Engineering will make every attempt to assist you in obtaining access by loaning this technology for use during the semester. Please be aware that supplies are limited, and we may not be able to fulfill all requests. If you need assistance securing technology (laptops, tablets, web cams, hotspots), please contact Theodore Caldwell, Assistant Dean for Equity and Inclusion, by email at tc@msu.edu.

Course Delivery Structure

This course will be delivered **in person**. You will need your MSU NetID to login to and get the course material from the *https://d2l.msu.edu/d2l/home/2523363* Students may forward their D2L email to an external email address (https://help.d2l.msu.edu/node/4410).

In D2L, you will access online lessons, course materials, homework assignments, and additional resources. Activities may consist of readings, discussion forums, email, and other online activities.

D2L Technical Assistance

If you need technical assistance at any time during the course or to report a problem you can:

- Visit the Distance Learning Services Support Site
- Visit the Desire2Learn Help Site (http://help.d2l.msu.edu/)
- Or call Distance Learning Services: (800) 500-1554 or (517) 355-2345

Resource Persons with Disabilities (RCPD)

- To make an appointment with a specialist, contact: (517) 353-9642
  or TTY: (517) 355-1293
- Web site for RCPD: http://MYProfile.rcpd.msu.edu

Part 2: Grading and Schedule

Graded Homework

Homework will be released in D2L. Homework is to be done independently unless otherwise discussed. Use of web or text resources is encouraged, but they must be cited.

Written homework problems provide exercise on concepts and methods. Some programming may be needed to complete homework. Programming is intended to reinforce algorithms and concepts and to get correct results; it is NOT intended to be an exercise in software development (so minimum documentation and usability but careful testing). Students will often use multiple tools or programs. A report of program/algorithm behavior and results is required. Programming work will be submitted electronically to https://d2l.msu.edu. **Please test how to submit a few days before the first homework deadline.** Unless otherwise stated, ANY programming language or computer can be used, while the Python language is preferred. Students will be responsible if conversion of data or program modules is needed. Students should seriously consider using Python due to its rapid development capability. Also, students should become familiar with one or more popular image processing tools. It should be clear when basic algorithms should be implemented by the student and not obtained from an existing library or function.

There are six homework assignments in total, as shown on the course calendar. HW0 is an ungraded assignment designed to help students set up the programming environment and become familiar with the submission process. Of the remaining five assignments, **three will be graded: HW2, HW4, and HW5, while HW1 and HW3 will be ungraded.** Although HW1 and HW3 do not directly contribute to the course grade, students are expected to work on them as practice, and material covered in these assignments may appear on the quizzes.

Late work will be accepted without penalty only with an acceptable excuse: without such, late penalty is 20% for the first day and another 20% for the next two days. No late work will be accepted three or more days after it's due. Some problems solved are far better than none. Each student has a total of 5 free late days for the semester, which can be used across assignments as needed.

Homework Report

The homework report should be a single .pdf or .doc file. Handwork or hand annotated images or code is acceptable and can be scanned into electronic form. Program modules, if requested, should be in tar or zip form.

Report format is not rigid, but should include the following elements.

- The problem[s] or homework objectives.
- The approach taken / major algorithm[s], possibly mentioning what alternatives were not taken.
- Results are most important, and *how* to get to the results is important too.
- Provide a few cases where things look good.
- Often there will be required cases.
- Document the parameters, process, etc.
- Provide a few cases where things don't look good.
- Document the parameters, process, etc.
- Concluding comments, how things went overall, what would be good to try if the work continued.

Quizzes

There will be three in-class quizzes during the semester. Each quiz will be announced one week in advance. The quizzes will be closed book.

Material from all homework assignments, including the ungraded HW1 and HW3, may be included in the quizzes.

Final Grade Computation

Grading is NOT competitive; individual performance will be used. Items will be weighted as follows:
3 homeworks (60%), 3 quizzes (30%), attendance (10%).

Grading Rubric

Range	Grade
(86, 100]	4.0
(79, 86]	3.5
(72, 79]	3.0
(65, 72]	2.5
(55, 65]	2.0
(48, 55]	1.5
(40, 48]	1.0
(00, 40]	0.0

Course Materials

Slides can be found in d2L. Slides are under constant improvement, but are never a substitute for class.

Course Calendar The following topics and readings are planned by week: reading assignments may be refined as we approach them.

Date	Topic	Assignments/ Things Due
Tue Sep 1	Intro to computer vision	
Thu Sep 3	Introduction & Projective Cameras I	**HW0 out**
Tue Sep 8	Projective Cameras II	
Thu Sep 10	Light and Shading	**HW1 Out**
Tue Sep 15	Numerical Linear Algebra Part I	
Thu Sep 17	Numerical Linear Algebra Part II	
Tue Sep 22	Linear and Nonlinear filtering	
Thu Sep 24	Detectors and Descriptors I	**HW2 Out**
Tue Sep 29	Detectors and Descriptors II	
Thu Oct 1	Transformations I	
Tue Oct 6	Transformations II	
Thu Oct 8	Linear Models	**HW2 Due, HW3 Out**
Tue Oct 13	Continuous Optimization	
Thu Oct 15	Backpropagation and Neural Nets	
Tue Oct 20	Convnets Part I	
Thu Oct 22	Convnets Part II	
Tue Oct 27	**No lecture (Fall break)**	
Thu Oct 29	Advanced Topic: ViT and VLM	**HW4 Out**
Tue Nov 3	Labeling Pixels	
Thu Nov 5	Detection	Quiz 2 (Tentative)
Tue Nov 10	Optical Flow	
Thu Nov 12	Tracking and Video Problems	
Tue Nov 17	Calibration and Intro to 3D	**HW4 Due, HW5 Out**
Thu Nov 19	Single-View Geometry	
Tue Nov 24	Epipolar Geometry	
Thu Nov 26	**No lecture (Thanksgiving)**	
Tue Dec 1	Stereo	
Thu Dec 3	3D Vision	Quiz 3 (Tentative)
Tue Dec 8	Structure from Motion	**HW5 Due**

Part 5: Course Policies

Participation

Students are expected to participate in all activities as listed on the course calendar.

Inform Your Instructor of Any Accommodations Needed

Michigan State University is committed to providing equal opportunity for participation in all programs, services and activities. Requests for accommodations by persons with disabilities may be made by contacting the Resource Center for Persons with Disabilities at 517-884-RCPD or on the web at rcpd.msu.edu. Once your eligibility for an accommodation has been determined, you will be issued a verified individual services accommodation ("VISA") form. Please present this form to me at the start of the term and/or two weeks prior to the accommodation date (test, project, etc). Requests received after this date will be honored whenever possible.

Drops and Adds

The last day to add this course is the end of the first week of classes. You should immediately make a copy of your amended schedule to verify that you have added or dropped this course. The details about the academic calendar can be found at https://reg.msu.edu/ROInfo/Calendar/Academic.aspx.

Commercialized Lecture Notes

Commercialization of lecture notes and university-provided course materials is not permitted in this course.

Complete Assignments

Assignments for this course will be submitted electronically through D2L unless otherwise instructed. Assignments must be submitted by the given deadline or special permission must be requested from instructor before the due date. Extensions will not be given beyond the next assignment except under extreme circumstances.

All discussion assignments must be completed by the assignment due date and time. Late or missing discussion assignments will affect the student's grade.

Academic Honesty:

The Department of CSE adheres to the policies on academic honesty as specified in General Student Regulations 1.0, Protection of Scholarship and Grades, and in the all-University Policy on Integrity of Scholarship and Grades, which are included in Spartan Life; Student Handbook and Resource Guide. Students who plagiarize will receive a 0.0 on the assignment/test/quiz.

Spartan Code of Honor academic pledge

"As a Spartan, I will strive to uphold values of the highest ethical standard. I will practice honesty in my work, foster honesty in my peers, and take pride in knowing that honor is worth more than grades. I will carry these values beyond my time as a student at Michigan State University, continuing the endeavor to build personal integrity in all that I do." (adopted by ASMSU on March 3, 2016, endorsed by Academic Governance on March 22, 2016, and recognized by the Provost, President, and Board of Trustees on April 15, 2016).

Tuition refund period Details can be found at https://reg.msu.edu/ROInfo/Calendar/Academic.aspx

Religious observance: Michigan State University has long had a policy to permit students, faculty/academic staff, and support staff to observe those holidays set aside by their chosen religious faith If you wish to be absent from class to observe a religious holiday, make arrangements in advance with the instructor.

Final Exam Policy can be found at https://reg.msu.edu/ROInfo/Calendar/FinalExam.aspx

The Office of Institutional Equity policies can be found at:

- Disability and Reasonable Accommodation Policy-https://oie.msu.edu/policies/disability-and-reasonable-accommodation-policy.html
- Policy on RVSM- https://oie.msu.edu/policies/rvsm.html

Tolerance and Civility- " MSU strives to build an academic community with living and learning environments that expects tolerance of viewpoints and civility toward others, whether at public forums, athletic events, in residential communities, classrooms or laboratories.
We call upon all who participate in university events to promote tolerance and civil behavior and to hold themselves to high standards that reflect the university's commitment to respect viewpoints that may be different from their own. Only by respecting individuals with diverse perspectives and ideas can we build an environment of civility that is conducive to advancing knowledge and transforming lives."